

MVLU COLLEGE

Aim: To set up and configure an open-source Intrusion Detection System (IDS) using Snort to monitor network traffic and detect potential security breaches or malicious activities.

1: System Update & Installation

[illegible]

2: Identifying the Network Interface

```
C:\sp08\BQ210>
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>nul
nul
C:\sp08\BQ210>cd /c/C:/Windows/System32/cd
C:\sp08\BQ210>$ short -v

-----
      *-* Short! c*-
****
Version 2.0.15.1 GRE (Build 15125) x86_64
By Martin Stuech & The Short Team: http://www.short.org/contact.htm
Copyright (C) 2004-2019 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2011 Sourcefire, Inc., et al.
Using libpcap version 1.10.6 (64-bit time_t, with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3.1

C:\sp08\BQ210>$ ls add-
ls: <IOERRACK_IP,IGMP,IP> mru 65536 qlisc noqueue state UNKNOWN group default qlen 1000
link/linkback 00:00:00:00:00:00 bnd 00:00:00:00:00:00
inet 172.20.0.1/8 scope host ls
    valid_lft forever preferred_lft forever
inet 16.255.255.255/32 bnd 16.255.255.255 scope global ls
    valid_lft forever preferred_lft forever
inets ::1/128 scope host proto kernel_ls
    valid_lft forever preferred_lft forever

C:\sp08\BQ210>$ ethb: <BROADCAST,MULTICAST,UP,LOWER_UP> mru 1500 qlisc mq state up group default qlen 1000
link/ether 08:15:5d:db:f2:36 bnd ff:ff:ff:ff:ff:ff
altname eno1334of214
inet 172.21.100.130/20 bnd 172.21.131.255 scope global ethb
    valid_lft forever preferred_lft forever
inets fe80::225:5dff:fe4b:f236/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever

C:\sp08\BQ210>$
```


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5: Live IDS Alert Output (ICMP Ping Detection)

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Aqib>ping 172.21.100.158

Pinging 172.21.100.158 with 32 bytes of data:
Reply from 172.21.100.158: bytes=32 time<1ms TTL=64
Reply from 172.21.100.158: bytes=32 time<1ms TTL=64
Reply from 172.21.100.158: bytes=32 time<1ms TTL=64
Reply from 172.21.100.158: bytes=32 time<1ms TTL=64

Ping statistics for 172.21.100.158:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Aqib>
```

```
sqib@A23B:~$
1 byte states : 1.82
2 byte states : 13.96
4 byte states : 0.00
-----
[ Number of patterns truncated to 20 bytes: 1010 ]
Snort DQ configured to passive.
Acquiring network traffic from "eth0".
Reload thread starting...
Reload thread started, thread 0x2814e932eb (574)
Decoding Ethernet

----- Initialization Complete -----

*% Snort! %*
Version 2.9.15.1 GME (Build 15125)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (64-bit time_t, with TPACKET_V3)
Using NDL version: 8.29 2016-06-14
Using DLIB version: 1.11.1

Rules Engine: SF_Snort_DETECTION_ENGINE Version 1.1 (Build 1)
Preprocessor Object: SF_Snort Version 1.1 (Build 4)
Preprocessor Object: SF_DUP Version 1.0 (Build 1)
Preprocessor Object: SF_REPUTATION Version 1.1 (Build 1)
Preprocessor Object: SF_NORMS Version 1.1 (Build 1)
Preprocessor Object: SF_PDP Version 1.0 (Build 1)
Preprocessor Object: SF_DCEPC2 Version 1.0 (Build 1)
Preprocessor Object: SF_Dns Version 1.1 (Build 4)
Preprocessor Object: SF_Sm Version 1.1 (Build 1)
Preprocessor Object: SF_FTPHUNT Version 1.2 (Build 1)
Preprocessor Object: SF_GTP Version 1.1 (Build 1)
Preprocessor Object: SF_Snort Version 1.1 (Build 0)
Preprocessor Object: SF_Sm Version 1.1 (Build 1)
Preprocessor Object: squid Version 1.1 (Build 5)
Preprocessor Object: SF_DNS Version 1.1 (Build 1)
Preprocessor Object: SF_SIP Version 1.1 (Build 1)

Commencing packet processing (pid=574)
00/24-05:33:38.550421 *** [1:1000001:1] ICMP Ping Detected *** [Priority: 0] [ICMP] 172.21.96.1 -> 172.21.100.158
00/24-05:33:38.550530 *** [1:1000001:1] ICMP Ping Detected *** [Priority: 0] [ICMP] 172.21.100.158 -> 172.21.96.1
00/24-05:33:31.550621 *** [1:1000001:1] ICMP Ping Detected *** [Priority: 0] [ICMP] 172.21.96.1 -> 172.21.100.158
00/24-05:33:31.550626 *** [1:1000001:1] ICMP Ping Detected *** [Priority: 0] [ICMP] 172.21.100.158 -> 172.21.96.1
00/24-05:33:32.560323 *** [1:1000001:1] ICMP Ping Detected *** [Priority: 0] [ICMP] 172.21.96.1 -> 172.21.100.158
00/24-05:33:32.560367 *** [1:1000001:1] ICMP Ping Detected *** [Priority: 0] [ICMP] 172.21.100.158 -> 172.21.96.1
00/24-05:33:33.570113 *** [1:1000001:1] ICMP Ping Detected *** [Priority: 0] [ICMP] 172.21.96.1 -> 172.21.100.158
00/24-05:33:33.570189 *** [1:1000001:1] ICMP Ping Detected *** [Priority: 0] [ICMP] 172.21.100.158 -> 172.21.96.1
```

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6: Nmap Port Scan & Detection Alert

```

%~> Smart! v+
Version 2.9.15.1 GM (Build 15126)
By Martin Rensch & The Smart Team: http://www.smart.org/contact.htm
Copyright (C) 2011-2015 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.08.4 (64-bit time_t, with TPACKET_V3)
Using NPM version: 6.20 2014-06-14
Using IDS version: 1.1.1

Rules Engine: SF_SMART_DETECTION_ENGINE Version 1.1 (build 1)
Progression Object: SF_SIGNATURE Version 1.1 (build 4x)
Progression Object: SF_SCAN Version 1.0 (build 1)
Progression Object: SF_REPUTATION Version 1.1 (build 1)
Progression Object: SF_METADATA Version 1.1 (build 1)
Progression Object: SF_DNS Version 1.0 (build 1)
Progression Object: SF_IPAPC2 Version 1.0 (build 1)
Progression Object: SF_IDS Version 1.1 (build 4x)
Progression Object: SF_SIP Version 1.1 (build 1)
Progression Object: SF_TTPOLICY Version 1.2 (build 13)
Progression Object: SF_CSP Version 1.1 (build 1)
Progression Object: SF_SWP Version 1.1 (build 4x)
Progression Object: SF_VM Version 1.1 (build 1)
Progression Object: smail Version 1.1 (build 1)
Progression Object: SF_DWP Version 1.1 (build 1)
Progression Object: SF_IDP Version 1.1 (build 1)

Running packet processing (144791)
[24-01-01-04-44.87829] [**] [1:1000002:1] Possible TCP SYN Scan Detected [**] [Priority: 0] [TCP] 172.21.96.1:59027 -> 172.21.100.150:80
[24-01-01-04-45.18186] [**] [1:1000002:1] Possible TCP SYN Scan Detected [**] [Priority: 0] [TCP] 172.21.96.1:59027 -> 172.21.100.150:80
[24-01-01-04-45.89468] [**] [1:1000002:1] Possible TCP SYN Scan Detected [**] [Priority: 0] [TCP] 172.21.96.1:59027 -> 172.21.100.150:80
[24-01-01-04-46.40712] [**] [1:1000002:1] Possible TCP SYN Scan Detected [**] [Priority: 0] [TCP] 172.21.96.1:59027 -> 172.21.100.150:80
[24-01-01-04-46.92188] [**] [1:1000002:1] Possible TCP SYN Scan Detected [**] [Priority: 0] [TCP] 172.21.96.1:59027 -> 172.21.100.150:80
[24-01-01-04-47.06793] [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.21.96.1 -> 172.21.100.150
[24-01-01-04-47.46709] [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.21.100.150 -> 172.21.96.1
[24-01-01-04-48.04310] [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.21.100.150 -> 172.21.96.1
[24-01-01-04-48.16810] [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.21.100.150 -> 172.21.96.1
[24-01-01-04-48.31772] [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.21.100.150 -> 172.21.96.1
[24-01-01-04-51.63187] [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.21.100.150 -> 172.21.96.1
[24-01-01-04-55.12618] [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.21.100.150 -> 172.21.96.1
[24-01-01-04-56.64106] [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] [ICMP] 172.21.100.150 -> 172.21.96.1

```

7: Wireshark Packet Capture

The screenshot displays a Wireshark interface capturing network traffic on the eth0 interface. The top section shows the packet list with columns for No., Time, Source, Destination, Protocol, Length, and Info. The selected packet is #19, an ICMP Echo request from 172.21.186.1 to 172.21.186.158.

No.	Time	Source	Destination	Protocol	Length	Info
16	26.478492200	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=133/43600, ttl=128 (reply in 16)
17	27.400312200	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) reply id=0x0001, seq=133/43600, ttl=64 (request in 25)
18	27.400555100	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=134/43604, ttl=128 (reply in 18)
19	28.500879100	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=135/43928, ttl=128 (reply in 20)
20	28.501899600	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) reply id=0x0001, seq=135/43928, ttl=64 (request in 19)
21	29.514442900	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=136/50176, ttl=128 (reply in 22)
22	29.515062500	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) reply id=0x0001, seq=136/50176, ttl=64 (request in 21)
192	526.540897500	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=137/50432, ttl=128 (reply in 193)
193	526.540917200	172.21.186.158	172.21.186.1	ICMP	74	Echo (ping) reply id=0x0001, seq=137/50432, ttl=64 (request in 192)
194	527.565344700	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=138/50688, ttl=128 (reply in 195)
195	527.566367100	172.21.186.158	172.21.186.1	ICMP	74	Echo (ping) reply id=0x0001, seq=138/50688, ttl=64 (request in 194)
196	528.574832500	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=139/50944, ttl=128 (reply in 197)
197	528.575765100	172.21.186.158	172.21.186.1	ICMP	74	Echo (ping) reply id=0x0001, seq=139/50944, ttl=64 (request in 196)
198	529.590820600	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=200/51200, ttl=128 (reply in 199)
199	529.591239900	172.21.186.158	172.21.186.1	ICMP	74	Echo (ping) reply id=0x0001, seq=200/51200, ttl=64 (request in 198)
272	830.670161000	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=201/51456, ttl=128 (reply in 273)
273	830.670931200	172.21.186.158	172.21.186.1	ICMP	74	Echo (ping) reply id=0x0001, seq=201/51456, ttl=64 (request in 272)
274	831.685137100	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=202/51712, ttl=128 (reply in 275)
275	831.686597100	172.21.186.158	172.21.186.1	ICMP	74	Echo (ping) reply id=0x0001, seq=202/51712, ttl=64 (request in 274)
636	832.690311100	172.21.186.1	172.21.186.158	ICMP	74	Echo (ping) request id=0x0001, seq=203/51968, ttl=128 (reply in 637)

The bottom pane shows the details of the selected packet (#19), categorized as Internet Control Message Protocol. It includes fields for Frame 15's Packet, Ethernet II, Microsoft Windows [00:15:5d:4c:f2:36], and Internet Protocol Version 4. The raw data field at the bottom displays the hexadecimal representation of the packet bytes.

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